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# C3 lowering in adult APP KI mice rescues synapses and spares cognitive decline

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## Abstract

**Background** We previously reported that germline complement C3 deletion protected cognition and hippocampal synapses in aged APP/PS1dE9 mice despite increased amyloid plaques. To assess whether global C3 lowering in adult amyloid mice might be neuroprotective, we crossed our C3 inducible conditional mice with *APP<sup>NL-G-F/NL-G-F</sup>* knockin mice.

**Methods** *C3fl/fl;Rosa26-Cre-ERT2* (C3iKO) mice were crossed with *C3fl/fl;APP<sup>NL-G-F/NL-G-F</sup>* mice to generate *APP;C3iKO* mice, which were treated with tamoxifen (TAM, n = 16) or corn oil (CO, n = 16) for 5 consecutive days at 3.6 months of age. Serum was collected 30 days post-treatment and at study termination to measure C3 levels. Behavioral testing was conducted at 15 months, followed by euthanasia and brain tissue analysis via ELISA, immunofluorescence, qPCR, and RNAseq.

**Results** Serum C3 levels were reduced by ~85% 30-days post-TAM treatment and ~70% at the end of the study. TAM-treated *APP;C3iKO* mice performed significantly better on cognitive tests compared to CO-treated mice. Complement C3 and C1q levels were significantly reduced in brain. No differences were observed in cerebral amyloid load. Iba-1 immunoreactivity of microglia was reduced in the hippocampal CA3 region of TAM-injected mice, while no differences were seen in GFAP labeling of astrocytes. However, hippocampal plaques were associated with fewer CD68-positive microglia and GFAP-positive astrocytes. Additionally, presynaptic markers (SYN and Bassoon) and postsynaptic markers (PSD95 and Homer1) were elevated in the CA3 and CA1 subregions of hippocampus. TAM treatment of *APP;C3iKO* mice led to reduced mRNA levels of *C3*, *TNF-α*, *IL-10*, *CX3CR1*, and *IL-6*. RNAseq identified 1071 differentially expressed genes (569 upregulated, 502 downregulated), with many related to synaptic signaling (e.g., Bassoon, Homer1, Syn2, SYNPO, and SNAP25).

**Conclusion** Global complement C3 lowering in adult *APP*-KI mice mitigated neuroinflammation, preserved synaptic integrity, and improved cognition despite having no effect on cerebral amyloid. These findings suggest that complement-targeted therapies, especially delivered to the brain, may protect synapses and slow cognitive decline.

**Keywords** *APP*-KI mice, Complement C3, Cognition, Amyloid-β, Hippocampal synapses, Neuroinflammation

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## Background

Neuroinflammation is a significant driving force for the onset and progression of Alzheimer's disease (AD), a progressive neurodegenerative disorder [1, 2]. The complement system is an essential player in the immune system, and its components are involved in neuroinflammatory reactions, pathogen defense, and elimination of cellular debris [3, 4]. The complement component 3 (C3) protein, an innate immune molecule, acts as a central hub for the classical, lectin, and alternative pathways of the complement system to coordinate downstream signaling pathways. In the classical complement pathway, the C1 complex, composed of C1q and serine proteases C1r and C1s, binds to antibody-antigen immune complexes causing a conformational change in C1q which activates the enzymatic activity of C1r and C1s. This leads to the cleavage of C4 and C2, generating C4b/C2b, known as C3 convertase, which cleaves C3 into C3a and C3b. C3b is subsequently cleaved into iC3b, which opsonizes pathogenic immune complexes and facilitates their binding to complement receptor 3 (CR3) on microglia, promoting phagocytosis of the immune complexes. C3b also binds immune complexes but binds complement receptor 1 (CR1) on erythrocytes and phagocytes, including microglia and macrophages. Further assembly of C3b into C5 convertase leads to the generation of the cleavage product C5b, which subsequently associates with C6, C7, C8, and multiple copies of C9 to form the membrane attack complex (MAC). MAC creates pores in the membranes of cell (e.g., pathogens), leading to their lysis [5–7]. In addition, C3 aids in maintaining the equilibrium between synaptic pruning and synapse formation and facilitates the elimination of inappropriate synapses to promote healthy brain development [8–10]. Several studies have indicated that dysregulation of the complement system in Alzheimer's is linked to the binding of complement components, such as C1q and C3 cleavage products to amyloid plaques, which can trigger an excessive neuroinflammatory response that may lead to neurodegeneration and cognitive decline [11, 12]. Chronic activation of C3 protein is linked to exacerbated microglia-dependent neurodegeneration and subsequent synaptic loss in the human AD brain and amyloid mouse models [13].

Alzheimer's disease (AD) is characterized by amyloid plaque formation, tau protein aggregation, neuroinflammation, synaptic deficits, and neuronal loss, all of which contribute to cognitive and memory decline [14, 15]. Complement proteins have emerged as important innate immune contributors because of their dual role in immunological function in the clearance of pathogens and the pathogenesis of Alzheimer's disease [4, 16]. Complement activation plays an important role

in the crosstalk between astrocytes and microglia to mediate a protective role in the early stages of AD; however, at later stages, it can lead to synaptic deficits and associated cognitive decline [17, 18]. Furthermore, the complement activation leads to cleavage of astrocyte-secreted C3, releasing C3a, which interacts with the microglial C3a receptor (C3aR), contributing to  $\beta$ -amyloid pathology in AD mouse models of AD [17, 19]. In recent years, the recognition of C3 involvement in AD pathology has increased [20, 21]. Our previous study demonstrated that despite an elevated A $\beta$  plaque load, germline C3-deficiency (C3KO) preserved hippocampal synapses, neuronal integrity, and cognitive function in both aging wild-type (WT) mice and APP/PS1dE9 mice [22]. Elevated C3 levels has been reported in mouse models of amyloidosis (PS2APP) and tau pathology (TauP301S), and C3 deficiency in these mice models led to protection against hippocampal synapse loss, suggesting that the inhibition of complement pathway may protect the brain against aging and AD [13]. Despite growing evidence suggesting that C3 inhibition throughout life protected against the progression of AD pathology in various mouse models, it remains unclear whether C3 suppression shortly after the onset of plaque deposition will protect against hippocampal synapse loss and associated cognitive decline in aging amyloid mice.

The APP-KI mouse model (*APP<sup>NL-G-F/NL-G-F</sup>*), an amyloid mouse model generated by knocking in *Swedish*, *Iberian*, and *Arctic* mutations in the human APP gene [23], develops early and robust deposition of cerebral amyloid plaques starting as early as 3 months of age, leading to amyloid-driven neuroinflammation and loss of synaptic transmission with aging [23, 24]. Because these mice express physiological levels of mutant human APP [23], we selected this model to cross with our global C3 inducible conditional knockout mouse model, *C3<sup>fl/fl</sup>; Rosa26-Cre-ERT2* (C3iKO) mice (described in our companion paper by Batista et al.) to generate *APP;C3iKO* mice. Our study was performed in parallel with a similar study by Dujardin et al. (Vandenbroucke lab; companion paper). Both studies demonstrated that global C3 lowering after the onset of A $\beta$  plaque deposition had no effect on cerebral plaque burden or overall gliosis. Here, we extend these studies by demonstrating that global C3 lowering in adult APP-KI mice reduced neuroinflammation, especially near plaques, and protected both hippocampal synapses and cognition despite not affecting the amyloid plaque load.

## Methods

### Animals

To generate *APP*;*C3iKO* mice for this study, *APP<sup>NL-G-F/NL-G-F</sup>* knock-in (*APP*-KI) mice supplied by Dr. Takaomi Saido, Riken Brain Institute, Japan) were first crossed with *C3* floxed (*C3<sup>fl/fl</sup>*) mice (chimeric floxed *C3* mice on a C57BL/6 J background) to obtain double heterozygous offspring (*APP<sup>+/-</sup>*;*C3<sup>fl/wt</sup>*). These double heterozygous mice were intercrossed to produce *APP* homozygous mice carrying the floxed *C3* allele (*APP<sup>+/+</sup>*;*C3<sup>fl/fl</sup>*). In parallel, *C3<sup>fl/fl</sup>* mice were cross-bred twice with *Rosa26-CreERT2<sup>+/+</sup>* to generate *C3<sup>fl/fl</sup>*;*Rosa26-CreERT2<sup>+/+</sup>* mice which were crossed with *C3<sup>fl/fl</sup>*;*APP<sup>+/+</sup>* mice to introduce the inducible Cre transgene. Offspring carrying the *C3<sup>fl/fl</sup>*, *APP<sup>+/+</sup>*, and *Rosa26-CreERT2<sup>+/+</sup>* alleles were selected as breeders to establish the *APP*;*C3iKO* line (*C3<sup>fl/fl</sup>*;*APP<sup>+/+</sup>*;*Rosa26-CreERT2<sup>+/+</sup>*), which was maintained through subsequent crossings with *APP<sup>+/+</sup>*;*C3<sup>fl/fl</sup>* mice. *APP*;*C3iKO* mice used in the present study were homozygous for the *APP<sup>NL-G-F</sup>* knock-in allele and are referred to throughout as *APP*;*C3iKO*. All mice were genotyped using ear snips to confirm the presence of *APP*-KI, *C3<sup>fl/fl</sup>*, and Cre transgenes. All mice were kept in a pathogen-free environment with a 12-h light/dark cycle and unlimited access to food and drinks. The Institutional Animal Care and Use Committee (IACUC) of Brigham and Women's Hospital approved the experimental procedures, which followed protocols aimed at minimizing animal suffering and the number of animals utilized.

### Tamoxifen and corn oil administration

Tamoxifen stock solution (20 mg/ml) was prepared by dissolving tamoxifen (Sigma-Aldrich, CAS # T5648-1) in corn oil at 37 °C overnight on a shaker and stored in a foil-wrapped tube because of its light-sensitive nature. Tamoxifen (TAM) was administered intraperitoneally (i.p.) at ~3.6 months of age to the TAM group of mice (*n* = 16) at a dose of 75 mg/kg body weight every 24 h for five consecutive days at 3.6 month of age. A one mL syringe fitted with a 25-gauge, 5/8" beveled needle (BD) was used to administer the injections. The injection site was sanitized using an alcohol pad before the injection to prevent infection. The mice were closely monitored throughout the injection and post-injection periods to check for signs of adverse effects. Mice in the CO group (*n* = 16) were injected with an equal volume of vehicle (CO) on the same schedule and were used as controls. A schematic diagram of the study design is provided in Fig. 1A.

### Baseline blood collection from saphenous vein

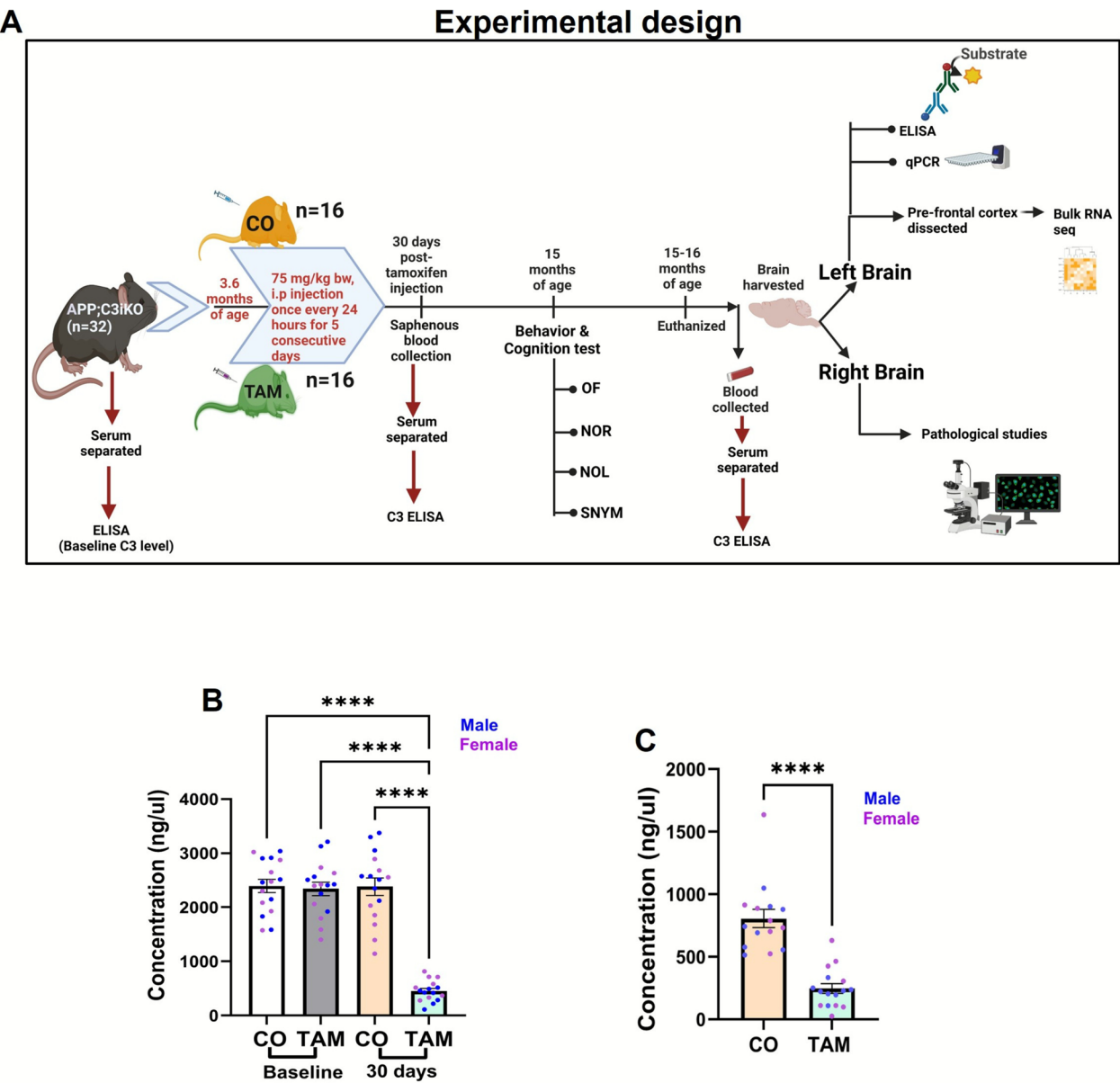
To check the level of C3 in the serum, we collected blood from the saphenous veins of the CO- and TAM-treated mice before the start of injections. Mice were restrained to expose the saphenous veins. The vein was punctured with a needle and 100–150 µL of blood was collected using a microhematocrit capillary tube. Tissue adhesion I (Vet-bond) was applied at the puncture site to prevent bleeding. Blood was centrifuged at 10,000 rpm for 10 min at 4 °C to separate the serum after allowing it to clot for 30–60 min at room temperature. After centrifugation, the serum was separated, transferred to a labeled microcentrifuge tube, and stored at –80 °C for further use.

### Behavioral and cognitive studies

The effect of C3 lowering in *APP*;*C3iKO* mice on behavioral and cognitive parameters, that is, locomotor activity, spatial memory, and recognition memory, was assessed using specific tests such as the Open Field (OF), Spatial Novelty Y-Maze (SNYM), Novel Object Recognition (NOR), and Novel Object Location. The Ethovision XT 14 computer-assisted video tracking software was used to record all behavioral and cognitive tests. Fifteen month-old *APP*;*C3iKO* mice from both the TAM- and CO-treated groups were subjected to behavioral and cognitive studies under controlled room conditions and at specific times between 10:00 a.m. and 5:00 p.m. to avoid variability. Prior to testing, all mice were handled for three consecutive days to minimize the stress and familiarize them with the experimenter, and each rodent was acclimatized to the experimental room for 30 min to minimize stress.

### Open field (OF)

The OF test was used to assess locomotor activity and anxiety in CO- and TAM- treated *APP*;*C3iKO* mice. The OF test consisted of a square-shaped plexiglass arena (50 cm long, 50 cm wide, and 40 cm high). At the beginning of the test, the animals were gently placed in the center of the arena, and their movements were recorded for a 10-min period using an overhead video camera connected to Ethovision automated tracking software. During the test, the distance traveled, and time spent in the center and peripheral arena were measured. Distance traveled corresponds to locomotor activity while time spent in the center is a measure of anxiolytic activity (i.e. the more time in the center, the less anxious the behavior). The arena was thoroughly cleaned with 70% ethanol before testing to avoid residual olfactory cues from previous mice.



### Serum C3 levels

**Fig. 1** Overview of the study and confirmation of C3 lowering. (A) The breeding scheme for the generation of *APP;C3iKO* mice. (B) The experimental design, outlining the timeline of tamoxifen (TAM) and corn oil (CO) injections at ~3.6 months of age and subsequent analyses. (C) Serum C3 levels were measured by ELISA at baseline (before the intraperitoneal injections) and 30 days post-TAM injection in *APP;C3iKO* mice, showing a significant reduction in serum C3 protein levels one month after TAM injections. (D) Serum C3 levels measured at the endpoint of the study (15–16 months of age). Data are presented as mean  $\pm$  SEM. Statistical analysis was performed using one-way ANOVA followed by Tukey's post hoc test, and pairwise comparisons were made using the Student's t-test. \*\*\*\* $p < 0.0001$ .  $N = 15\text{--}16$  mice per group

### Spatial novelty Y-maze (SNYM)

Spatial memory was assessed in mice using the SNYM protocol. The Y-maze consisted of a start arm, left arm, and right arm, all aligned directly below the camera. Each mouse underwent a 3-min habituation phase followed by

a 2-min intertrial interval (ITI), followed by a 3-min test phase. During habituation, the left and right arms were randomly blocked, and the blocked arm was considered novel. The test phase began when the mouse entered the left or right arm, and EthoVision software tracked

and analyzed the movement of the mice. The Y maze was cleaned with 70% ethanol and dried between mice. Data from the habituation and test phases were exported to Excel. The data were then analyzed, and the percent time spent in the novel arm in CO- and TAM-injected *APP*; *C3iKO* mice was calculated.

#### **Novel object recognition (NOR)**

The NOR test was used to evaluate recognition memory in mice by utilizing their innate preference for exploring novel objects rather than familiar objects. The NOR apparatus consisted of four black Plexiglas chambers (50 cm × 50 cm × 40 cm), with an overhead camera connected to the EthoVision XT 14 software, which facilitated video tracking of mouse interactions with objects. The protocol spanned three days. On day 1, the habituation phase, mice were acclimated to an open arena box for 10 min. On day 2, the training phase, each mouse underwent 10-min training sessions, exploring two identical objects (Legos or blocks). Day 3, the testing phase, each mouse was again familiarized with the two identical objects and after 60 min, one familiar object was replaced with a novel object, and the mouse was allowed to freely explore the objects for 10 min. Before each session, the chambers and objects were thoroughly cleaned with 70% ethanol and then dried. The objects were placed at marked positions to ensure consistency across the trials. Data from the EthoVision system were exported to an Excel file and analyzed to calculate the discrimination index (DI), which was determined as (time spent exploring the novel object – time spent exploring the familiar object)/(total time spent exploring both objects). The total exploration time per mouse is provided in the Supplemental Materials (Table 3).

#### **Novel object location (NOL)**

The NOL test was used to evaluate mouse recognition memory, exploiting their propensity to explore an object relocated to a novel position. The apparatus used for NOL was the same as that used for NOR. The test protocol consisted of three consecutive days. On Day 1, the habituation phase, mice were acclimated to the open arena chamber box for 10 min. On day 2, the training phase, each mouse underwent a 10-min training session, exploring two identical objects (Legos or blocks) placed at a fixed position. On day 3, the testing phase, each mouse was again familiarized with both objects at the same position; however, after a 60-min interval, one object was relocated to a new position while the second object remained in an initial position. Each mouse was placed at the center of the arena for a 10-min test session. Before each session, the chambers and objects were thoroughly cleaned with 70% ethanol and then dried. Data

from the EthoVision system were exported to an Excel file and analyzed to calculate the discrimination index (DI), which was determined as (time spent exploring the novel object – time spent exploring the familiar object)/(total time spent exploring both objects). The total exploration time per mouse is provided in the Supplemental Materials (Table 4).

#### **Brain extraction and tissue processing for immunohistochemistry**

Mice were euthanized with carbon dioxide and blood was collected by cardiac puncture from the right ventricle after bilateral thoracotomy. Following transcardial perfusion with ice-cold phosphate-buffered saline (PBS) to clear the blood, the brains were rapidly harvested and separated into left and right hemibrains. The left hemibrain was snap-frozen on dry ice and stored at –80 °C for biochemical and molecular studies, while the right hemibrain was fixed overnight at 4 °C in 4% paraformaldehyde and cryoprotected using sucrose gradients of 10%, 20%, and 30% until they settled at the bottom. A sliding microtome (Epreidia, HM450) was used to cut sagittal sections of the right hemibrain at a thickness of 30 µm, which were then stored in PBS containing 0.2% sodium azide at 4 °C for free-floating immunohistochemistry.

#### **C3 and C1q ELISA**

A 96-well high-binding microplate (Greiner Bio-One MICROLON 600) coated with 100 µL of goat anti-mouse C3 antibody (1:500 dilution, 4 µg/mL, M.P Biomolecules #55,463) in PBS or rabbit monoclonal anti-C1q antibody (Abcam, catalog AB 182451) was incubated overnight at 4 °C. The plate was then washed three times with PBS-Tween (0.05%) and the excess liquid was removed. A blocking solution of 100 µL 4% BSA in PBS was added to each well and incubated at 37 °C for 1 h. Samples were prepared in 4% BSA with dilutions of 1:50,000 for serum and 1:15 for T-per-soluble brain samples for the C3 ELISA, and for the C1q ELISA, T-per brain samples were diluted 1:200. Samples and standard serial dilutions were transferred to the plate containing blocking solution and incubated for 2 h at room temperature. The plate was washed with PBS-Tween (0.05%) three times and then 100 µL 1:1,000 diluted biotinylated detection antibody was added, incubated for 1 h at 37 °C, and washed again with PBS-Tween (0.05%) three times. Next, 100 µL 1:10,000 diluted streptavidin-HRP antibody in 4% BSA was added, incubated for 30 min at 37 °C, and washed three times with PBS-Tween (0.05%). Lastly, 100 µL TMB substrate (Thermo Scientific # 34,028) was added to each well and the color change was monitored until the

reaction was halted with 100  $\mu$ L of stop solution (Thermo Scientific # N600), and the plate was read at 450 nm.

#### Preparation of brain T-per soluble and GuHCl insoluble fractions for A $\beta$ ELISA

Snap-frozen tissue samples were weighed and mixed with five volumes of T-Per buffer (Thermo Fischer Scientific) containing a protease inhibitor cocktail (Calbiochem, Cat# 539,131) and phosphatase inhibitor cocktail (Calbiochem, Cat# 524,627), homogenized for 10 s, sonicated, vortexed, and ultracentrifuged at 100,000 $\times$ g for 60 min at 4 °C. The supernatant (T-per soluble) was collected and stored at -80 °C. The pellet obtained was homogenized in 5 M guanidine hydrochloride (GuHCl) (150 mg/mL, pH 8.0) and ultracentrifuged after 3–4 h incubation. The supernatant (GuHCl-insoluble fraction) was collected for the insoluble A $\beta$  ELISA and stored at -80 °C.

#### A $\beta$ ELISA

To measure insoluble A $\beta$  levels in the GuHCl-brain homogenate, a Meso Scale Discovery (MSD) 96-well multi-spot Human/Rodent (4G8) A $\beta$  Triplex Ultra-Sensitive Assay was used. The well plate was incubated in 1% Blocker A solution and placed on the shaker for an hour to block the capture antibodies against A $\beta$ x-38, x-40, and x-42 on the well plate and washed with 1 $\times$ Tris wash buffer three times. Further, 25  $\mu$ L of detection antibody (4G8) solution and 25  $\mu$ L of either sample or calibrators were added to the wells and incubated for 2 h at room temperature. After three rounds of washing with 1 $\times$ Tris Wash Buffer, 150  $\mu$ L of 2 $\times$ MSD Read Buffer T was added per well. Read Buffer was applied, and the plate was read on an MSD Sector Imager.

#### DAB (3,3'-Diaminobenzidine) and immunofluorescence immunostaining

Immunohistochemistry was performed on sagittal sections from the right hemibrain (CO=15 and TAM=16), as previously described [22]. Three to four sections per mouse were incubated with primary antibody overnight at 4 °C (see Supplemental Table 1 for a list of all antibodies). The sections were washed with PBS-T, incubated with biotinylated secondary antibodies, washed with PBS-T, and incubated with Vector ELITE ABC kits (Vector Laboratories) and developed with 3,3-diaminobenzidine (Sigma-Aldrich). For immunofluorescent staining, immunofluorescently-tagged secondary antibodies were incubated for 2 h. Sections were washed three times with PBS and coverslipped with mounting media (Vector) for DAB stained sections or ProLong Gold Antifade mounting media (Thermo Fischer Scientific) for immunofluorescently labeled sections. Omission of primary antibody (secondary alone) was used to evaluate the specificity of

the immunolabelling. Immunolabelling was not observed in the negative control sections.

#### Imaging and quantification

Images of GFAP, IBA1, S97, ThioS, and triple-immunolabelling (IBA1 + S97 + CD68) were captured using a Zeiss microscope Axioscan 7 slide scanner (Oberkochen, Germany). Each imaging session was captured using constant brightness, illumination intensity, and exposure time settings to ensure consistency. Captured images were analyzed, and the percentage of immunoreactivity in the hippocampus was calculated using ImageJ macros with the computer-assisted software ImageJ Fiji [25]. Confocal Z-stack images of triple immunofluorescent labelling (S97 + IBA1 + CD68), and double immunofluorescent labeling including S97 + GFAP, GFAP + C3, SYN + PSD 95, Bassoon + Homer1, and C3 + Bassoon were captured by spinning disk confocal microscopy (Nikon Eclipse T microscope with a Hamamatsu digital camera C10600 ORCA) using 40 $\times$  and 60 $\times$  objectives and visualized using Volocity software. Three images from each of three sections per mouse were captured from three equidistant planes 500  $\mu$ m apart and the confocal images were analyzed, and the percentage of co-labeling was calculated using an ImageJ macro. Synaptic puncta (SYN + PSD95) and (BSN + HOMER1) were quantified using Imaris $\times$ 64 software using the “Colocalization spots” X tension (oxinst.com) with a defined distance  $\leq$  threshold value of 5, which identifies puncta showing consistent overlap of pre- and postsynaptic markers.

#### RNA isolation and quantitative real-time PCR (qRT-PCR) reaction

The RNeasy Micro kit (Qualigen, Inc.), and its protocol were used for the extraction of total RNA from the prefrontal cortex of the brains of CO and TAM-injected *APP;C3i*KO mice. The concentration and purity of the extracted total RNA were determined using a Nanodrop spectrophotometer (Nanodrop Technologies, ND-1000) by measuring the absorbance at an A260/A280 ratio between 1.8 and 2.0, indicating high purity.

cDNA was synthesized using the iScript cDNA synthesis kit (Bio-Rad Laboratories) following the manufacturer's protocol. The total reaction volume was 20  $\mu$ L for cDNA synthesis from 1  $\mu$ g of isolated RNA. The reaction volume consists of 10  $\mu$ L of SYBR Green Master Mix (Applied Biosystems), 1.5  $\mu$ L of each 100  $\mu$ M forward and reverse primer (Supplemental Table 2), 0.5  $\mu$ L cDNA template, and 20  $\mu$ L nuclease-free water was added to each reaction. qRT-PCR was performed using a Quant Studio 5 Real-Time PCR System (Thermo Fisher Scientific). The cycling conditions were as follows: initial denaturation at 95 °C for 2 min followed by 40 cycles of

denaturation at 95 °C for 15 s, annealing at 60 °C for 30 s, and extension at 72 °C for 30 s. To confirm the specificity of the PCR products, a melting curve analysis was performed. The housekeeping gene  $\beta$ -actin was used to normalize gene expression, and the relative expression was calculated using the  $2^{-\Delta\Delta C_t}$  method. All samples were run in duplicate to ensure accuracy and reproducibility.

### Bulk RNA seq

Total RNA was extracted from the prefrontal cortex of 15 CO mice and 16 TAM mice using the RNeasy Mini Kit (Qiagen). RNA was further processed for sequencing using the Gene Expression Core Facility at the Broad Institute (MIT). Sequencing data in the form of FASTq files were further processed using Galaxy (usegalaxy.org) online software. Raw reads were subjected to quality control using FastQC, adapter and low-quality base trimming using trimmomatic [26], and alignment to the mouse (*Mus musculus*:mm9) reference genome with HISAT2. Feature counts were used to quantify aligned reads. Normalization and initial data processing were performed using DESeq2 on a Galaxy platform [27]. Differential expression analysis was performed using voom-limma [28]. Specifically, the read counts were normalized, converted with the voom function to log2-counts per million (log-CPM), and then fitted into a linear model. Heat maps and volcano plots were generated for visualization using an SRplot [29].

## Results

### Validation of global C3 lowering in *APP*;*C3iKO* mice

To confirm the efficacy of global complement C3 reduction in our *APP*;*C3iKO* mouse model, we conducted assessments 30 days after tamoxifen treatment. Blood samples were collected from the saphenous vein, two days before CO and TAM injections to quantify serum C3 levels at baseline. Thirty days after the last tamoxifen injection, blood was collected, and a C3 ELISA was performed on serum samples to quantify C3 levels. While there was no difference in serum C3 levels at baseline, serum C3 levels were significantly decreased (~85%) in TAM-treated *APP*;*C3iKO* mice compared to CO-injected mice ( $p < 0.0001$ ) 30 days after treatment, validating the successful suppression of C3 expression one month after TAM injections (Fig. 1B). Serum C3 levels remained low in 15–16 month-old TAM-treated versus CO-treated mice at end of the study (Fig. 1C), indicating sustained C3 lowering following TAM-treatment at 3.6 months of age.

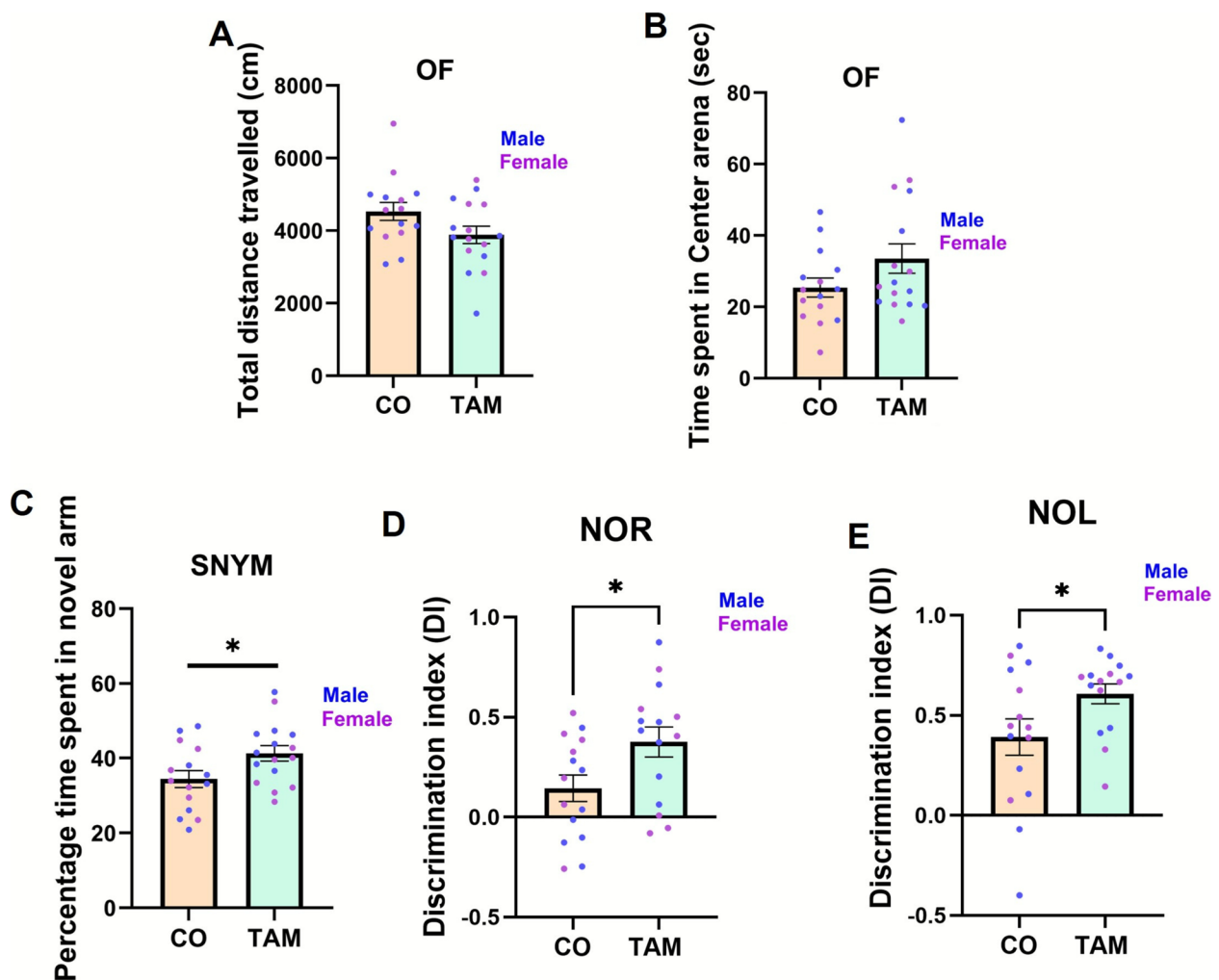
### C3 lowering at the onset of plaque deposition attenuated cognitive decline in *APP*;*C3iKO* mice at later ages

We conducted comprehensive behavioral and cognitive assessments of TAM- and CO-treated *APP*;*C3iKO* mice at 15 months of age to evaluate the impact of early complement C3 lowering on behavior and cognitive function at later ages. Behavioral testing included the OF test to assess locomotor activity and anxiety behavior, whereas cognitive function was evaluated through three distinct assessments: Spatial Novelty Y-Maze (SNYM), Novel Object Recognition (NOR), and Novel Object Location (NOL). TAM-treated *APP*;*C3iKO* mice exhibited a slight trend towards reduced total distance traveled compared to CO-injected mice (Fig. 2A), along with a tendency for increased time spent in the central arena (Fig. 2B). Although these differences were not statistically significant, the observed trends suggest a potential reduction in anxiety-like behavior in TAM-treated *APP*;*C3iKO* mice relative to their CO-treated *APP*;*C3iKO* counterparts.

In contrast, cognitive assessments revealed significant differences between CO- and TAM-treated *APP*;*C3iKO* mice. TAM-treated mice exhibited significantly preserved cognitive function in all three cognitive tests. In the SNYM test, which measures spatial working memory, TAM-injected *APP*;*C3iKO* mice displayed significantly ( $p = 0.032$ ) better performance characterized by increased time spent in the novel arm compared to CO-treated *APP*;*C3iKO* mice (Fig. 2C). Similarly, in the NOR test, TAM-treated *APP*;*C3iKO* mice demonstrated better recognition memory, exhibiting a significant higher discrimination index ( $p = 0.028$ ) compared to CO-injected treated *APP*;*C3iKO* mice (Fig. 2D). Furthermore, in the NOL test, which assesses spatial memory and object location memory, TAM-treated *APP*;*C3iKO* mice showed a significantly higher discrimination index ( $p = 0.046$ ), indicating superior performance compared to CO-treated mice (Fig. 2E).

### Brain C1q and C3 levels were reduced in aged *APP*;*C3iKO* mice following TAM-treatment

In our investigation of the long-term impact of lowering complement C3 in *APP*;*C3iKO* mice, we examined the levels of C1q and C3 by ELISA in T-per brain homogenate of *APP*;*C3iKO* mice at 15–16 months of age. We found a persistent, significant reduction in the levels of C3 in the brain tissues of TAM-treated mice compared to CO-treated mice ( $p = 0.0001$ ) as shown in Fig. 3A. Furthermore, we conducted double immunofluorescent staining of C3 with GFAP, as astrocytes (GFAP) are a major source of C3 in the brain. Both qualitative and quantitative data analysis indicated that GFAP-positive astrocytes expressed significantly less C3 ( $p = 0.0016$ ) in



**Fig. 2** C3 lowering spared cognitive decline at 15 months of age. (A) Open Field test – Total distance traveled. (B) Open Field test – Time spent in the center arena. (C) Spatial Novelty Y Maze test (SNYM). (D) Novel Object Recognition (NOR). (E) Novel Object Location (NOL). N = 15–16 mice/group. Data are presented as mean ± SEM. Statistical analysis was performed using pairwise comparisons with Student's t-test, and p values less than 0.05 were considered significant (\*p < 0.05)

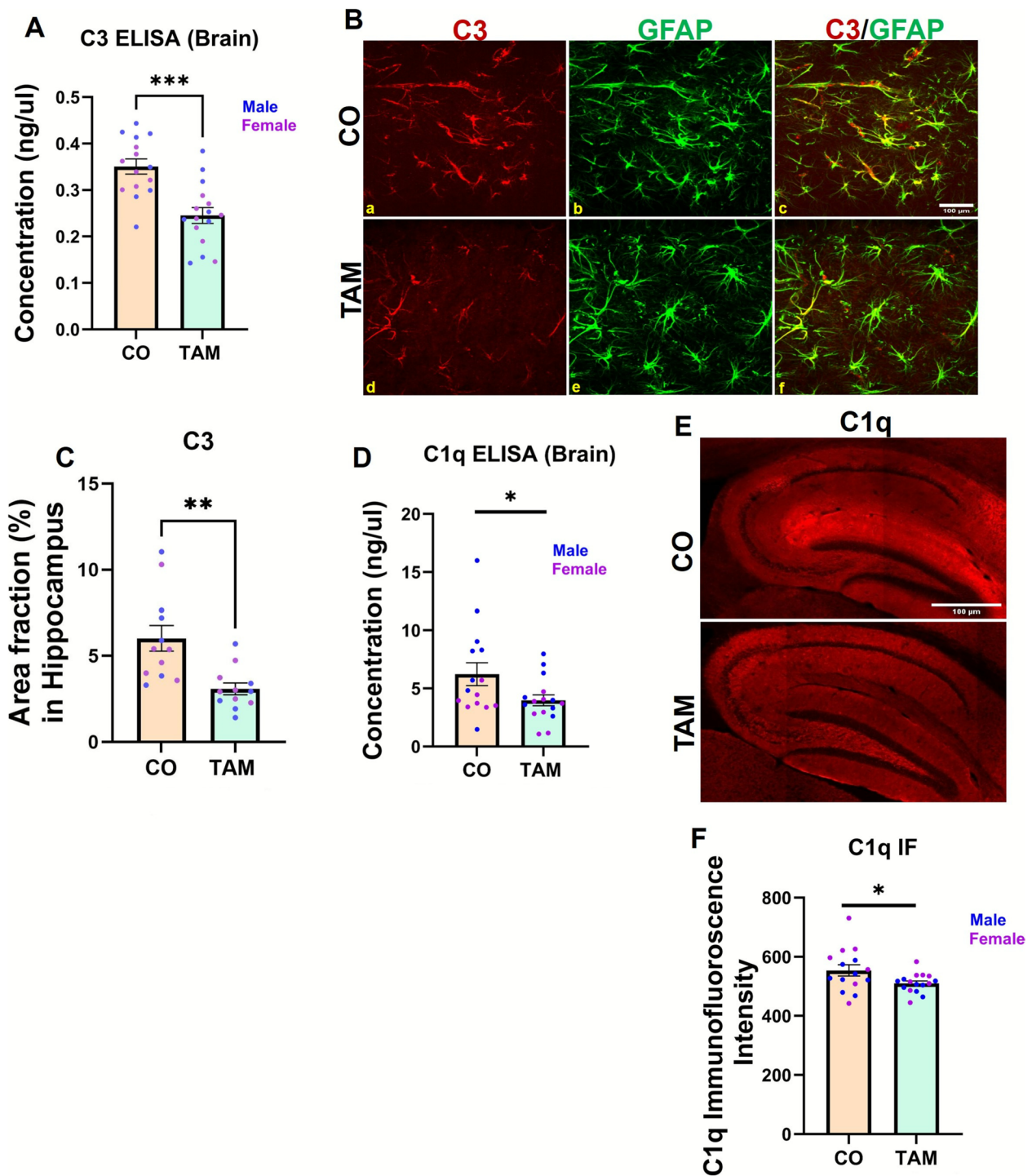
the hippocampi of TAM-treated mice compared to CO-treated mice (Fig. 3B and C).

In addition to the analysis of C3, we also assessed the levels of C1q, another key protein in the complement cascade. Brain C1q levels were significantly decreased ( $p=0.04$ ) by ELISA in TAM-treated *APP;C3iKO* mice compared to CO-treated control mice (Fig. 3D). C1q immunofluorescent staining demonstrated a significantly lower mean fluorescence intensity in the hippocampus of TAM-treated *APP;C3iKO* mice compared to CO-treated mice, indicating a significant reduction ( $p=0.0392$ ) in C1q levels in the brains of TAM-treated mice (Fig. 3E and F). Thus, both the biochemical assays and pathological examination demonstrated a concurrent reduction

in C1q levels along with C3 reduction after the onset of plaque deposition in the brains of TAM-treated mice.

#### Microglial immunoreactivity was reduced in hippocampal CA3, while astrocytes appeared unaffected by C3 lowering in *APP;C3iKO* mice

To characterize the effects of C3 lowering on microglia and astrocytes, we immunolabeled Iba-1 and GFAP, respectively, in the brains of TAM- and CO-treated *APP;C3iKO* mice. TAM-treated *APP;C3iKO* mice exhibited significantly less Iba-1 immunoreactivity in the CA3 hippocampal subregion compared to CO-treated control mice ( $p=0.0301$ ) (Fig. 4A & C). Notably, the hippocampal CA1 and dentate gyrus (DG) subregions showed no discernible differences



**Fig. 3** Lowering of brain C3 and C1q levels in *APP;C3iKO* mice at 15–16 months of age. (A) C3 protein levels in T-per brain homogenates was measured by ELISA at 15–16 months of age. (B) Representative double-immunofluorescence images showing a reduction in C3 (red) and GFAP (green) expression in TAM-injected *APP;C3iKO* mice. Scale bar = 100  $\mu$ m. (C) Quantitative analysis of C3 positive staining expressed as area of fraction in hippocampus region. (D) C1q levels was observed in T-per brain homogenates by C1q ELISA. (E) Representative images showing C1q expression in the hippocampus of *APP;C3iKO* mice. Scale bar = 100  $\mu$ m. (F) Quantitative analysis of C1q-positive staining expressed as mean fluorescence intensity. N = 15–16 mice/group. Data are presented as mean  $\pm$  SEM. Statistical analysis was performed using pairwise comparisons with Student's t-test, and p values less than 0.05 (\* $p < 0.05$ ) and 0.01 (\*\* $p < 0.01$ ) were considered significant

in microglial reactivity between treatment groups (Fig. 4A and D-E). Across the entire hippocampus, we observed a non-significant trend towards decreased Iba-1 immunoreactivity in TAM-treated *APP;C3iKO* mice compared to CO-treated control mice (Fig. 4A & B). No significant differences were observed in hippocampal GFAP immunoreactivity (Fig. 4F-J). These findings suggest a region-specific response of microglia within the hippocampus to C3 lowering in *APP;C3iKO* mice. Although the CA3 subregion exhibited significant alterations in microglial reactivity, the CA1 and DG subregions remained relatively unchanged. The lack of significant changes in astrocytic responses further underscores the specificity of observed microglial reactivity patterns in response to C3 modulation.

#### C3 lowering had no effect cerebral amyloid plaque load in *APP;C3iKO* mice

To determine the effect of early C3 suppression on amyloid-beta (A $\beta$ ) plaque deposition later in life, immunolabelling with a polyclonal anti-A $\beta$  antibody was used to quantify A $\beta$  plaque deposition in the hippocampus and its subregions (CA1, CA3, and DG) (Supp. Figure 1) at 15–16 months of age (Fig. 5A–B). We observed no significant difference in plaque deposition between TAM-treated and CO-treated *APP;C3iKO* mice. Thioflavin-S staining was performed to quantify fibrillar amyloid plaques. Similar to the S97 immunolabelling results, no significant difference was observed in fibrillar amyloid plaque deposition between the experimental groups (Fig. 5A and D). Insoluble A $\beta$  levels in the GuHCl brain homogenates were quantified by ELISA. Similar to the pathological data, there was no significant difference in insoluble A $\beta$  levels between TAM- and CO-treated *APP;C3iKO* mice (Fig. 5C). Together, these data indicate that C3 lowering at the onset of plaque deposition (3.6 months of age) had no effect on plaque deposition at 15–16 months of age in *APP;C3iKO* mice.

#### C3 lowering attenuated plaque-associated microglial phagocytosis and astrocytes in hippocampi of *APP;C3iKO* mice

Although C3 lowering had no effect on plaque deposition, we asked whether it might affect plaque-associated gliosis as seen in our previous study in germline C3KO in *APP/PS1* mice [22]. Triple immunofluorescent labeling of Ab (S97), microglia (Iba-1) and phagocytes (CD68) showed qualitatively less CD68-positive microglia associated with plaques in TAM-treated mice compared to CO-treated controls (Fig. 6A). Quantification of A $\beta$  plaque (S97)-associated Iba-1, a microglial/macrophage marker, showed no difference in TAM-treated mice compared to CO-treated control mice (Fig. 6B). Quantitative analysis of A $\beta$  plaque-associated CD68, a lysosomal marker for phagocytic microglia, revealed a significant decrease ( $p=0.044$ ) in A $\beta$ -associated CD68-positive microglia in TAM-injected mice compared to that in the CO-injected groups (Fig. 6C), indicating a reduction in microglial phagocytic activity following C3 lowering in TAM-injected *APP;C3iKO* mice, despite no change in overall microglial immunoreactivity. Overall, our results suggest a decrease in the neuroinflammatory response toward A $\beta$  plaques following C3 lowering in *APP;C3iKO* mice.

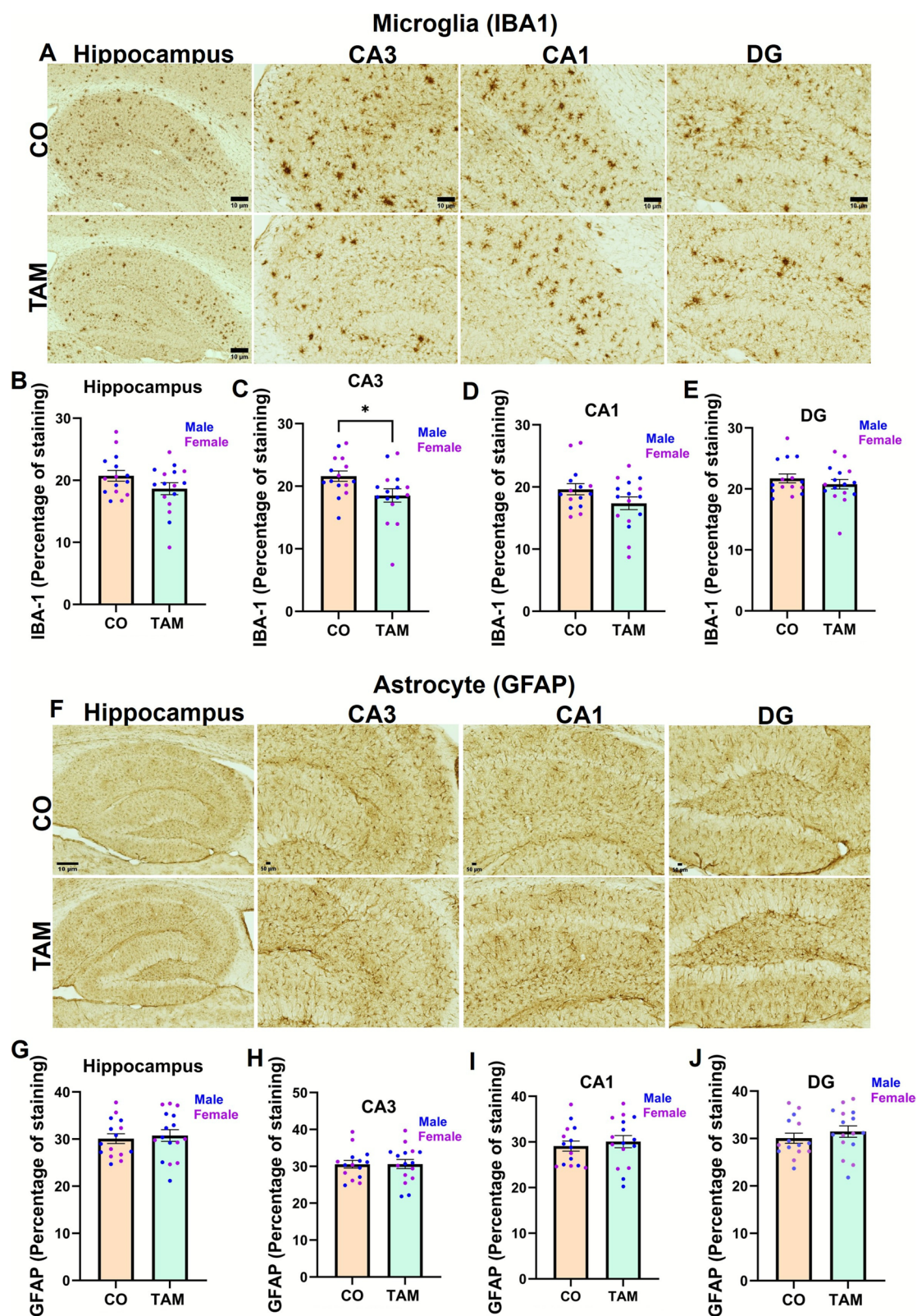
We also investigated the effect of complement component 3 (C3) reduction on plaque-associated astrocytes in the hippocampus of *APP;C3iKO* mice. Immunohistochemical analysis was performed on brain sections of TAM- and CO-treated *APP;C3iKO* mouse brains, focusing on the astrocyte marker glial fibrillary acidic protein (GFAP) and amyloid plaque marker (S97) (Fig. 6D). The density of GFAP-positive astrocytes surrounding amyloid plaques in the hippocampus was significantly reduced ( $P=0.0373$ ) in TAM-injected mice (Fig. 6E). This indicates that the recruitment of activated astrocytes near amyloid deposits was decreased by the C3 reduction.

#### Early C3 lowering impacted gene expression of immune genes in the brains of *APP;C3iKO* mice at 15–16 months of age

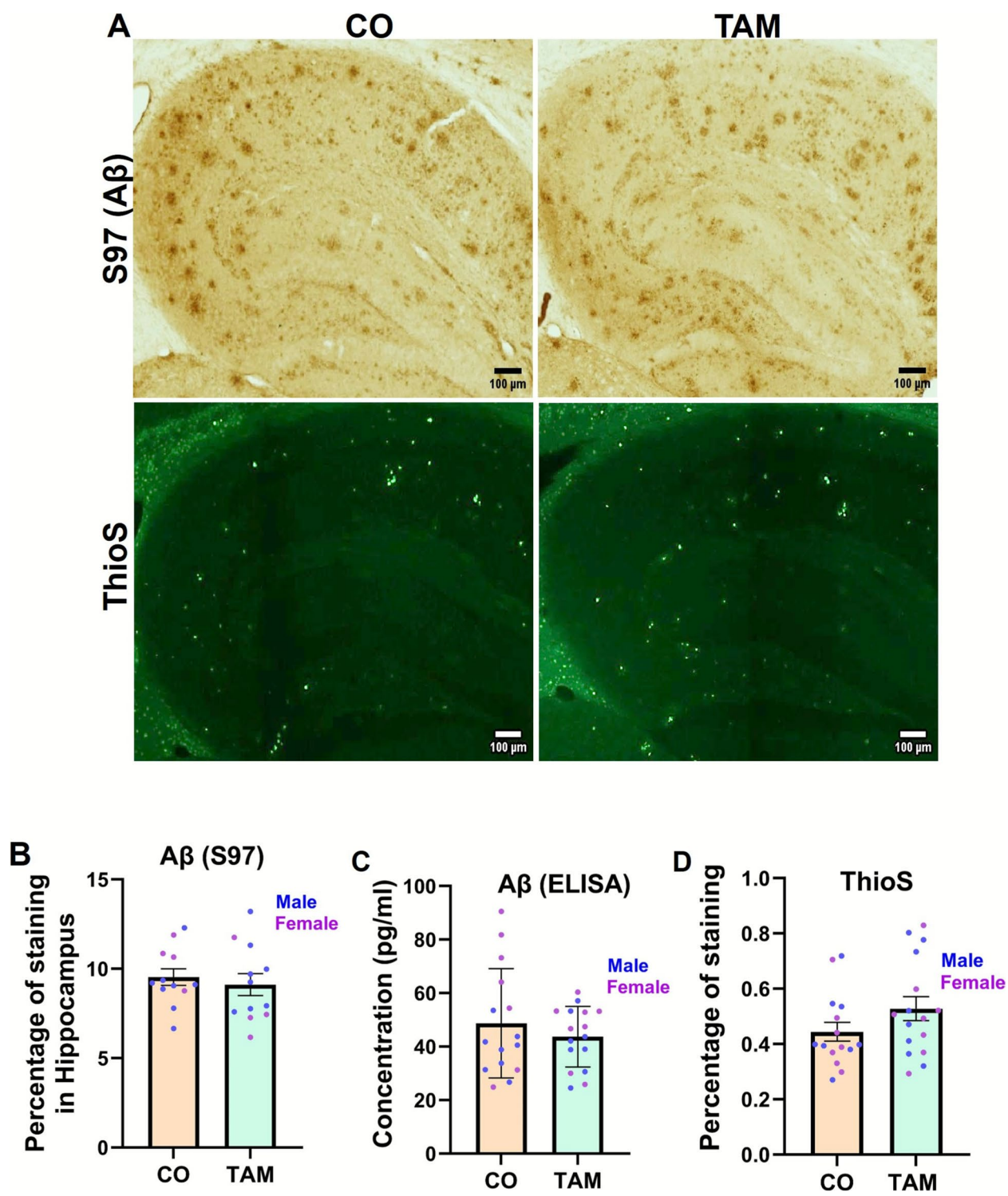
We conducted qPCR to examine the effect of C3 reduction on mRNA levels of *C1q*, proinflammatory cytokines (*TNF $\alpha$* , *IL-6*, *IL-1 $\beta$* ) chemokine receptor *CXCR1*, and

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**Fig. 4** Hippocampal microglial and astrocyte immunoreactivity at 15–16 months of age. (A) Representative immunostained images of the microglial marker Iba-1 in the hippocampus and its subregions (CA3, CA1, and DG). Scale bar = 10  $\mu$ m. (B–E) Quantitative analysis of Iba-1 expression in the hippocampus, CA3, CA1, and DG. (F) Representative images of the astrocytic marker GFAP in the hippocampus, CA3, CA1, and DG of *APP;C3iKO* mice at 15–16 months of age. (G–J) Quantitative analysis of GFAP expression in the hippocampus, CA3, CA1, and DG. Scale bars = 10  $\mu$ m and 50  $\mu$ m. N = 15–16 mice/group. Data are presented as mean  $\pm$  SEM. Statistical analysis was performed using pairwise comparisons with Student's t-test, and p-values less than 0.05 (\* $p < 0.05$ ) were considered significant



**Fig. 4** (See legend on previous page.)



**Fig. 5** Aβ plaque burden in *APP;C3iKO* mice. (A) Representative images of general Aβ (S97) and fibrillar amyloid (Thioflavin S) staining in the hippocampus of *APP;C3iKO* mice at 15–16 months of age. (B) Quantification of Aβ (S97) staining, expressed as percent area fraction. (C) Aβ levels measured by ELISA in GHCI brain homogenates at 15–16 months of age, showing a non-significant decrease in Aβ levels in TAM-injected mice compared to CO-injected mice. (D) Quantification of ThioS staining, expressed as the area fraction (percentage of staining), showing no significant change in fibrillar plaque load. Scale bar = 100 μm. N = 15–16 mice/group. Data are presented as mean ± SEM. Statistical analysis was performed using pairwise comparisons with Student's t-test

complement receptor *CR3* (*CD11b*, *CD18*). *C3* mRNA levels were significantly reduced in *APP*;C3iKO mouse brain at 15–16 months of age (Fig. 7A). This confirms sustained reduction in *C3* expression after TAM treatment in *APP*;C3iKO mice. No notable changes were observed in *C1qa* mRNA, although *C1qb* and *C1qc* mRNAs were modestly reduced but did not reach statistical significance in TAM-treated *APP*;C3iKO mice (Fig. B-D). The mRNA expression levels of proinflammatory and anti-inflammatory cytokines *TNF $\alpha$*  ( $p=0.0192$ ) and *IL-10* ( $p=0.015$ ), respectively, were significantly reduced in TAM-treated *APP*;C3iKO mice (Fig. 7E & I). The reduction in *TNF $\alpha$*  mRNA indicates reduced neuroinflammation in *APP*;C3iKO mice with *C3* lowering. Additionally, pro-inflammatory *IL-6* mRNA levels showed a strong, but non-significant trend for reduction in TAM-treated mice ( $p=0.065$ ; Fig. 7F). This suggests a tendency toward less inflammation due to *C3* lowering in *APP*;C3iKO mice. There was no significant change or trend in the mRNA levels of the proinflammatory cytokine *IL-1 $\beta$*  mRNA (Fig. 7G). Additionally, TAM-injected *APP*;C3iKO mice showed markedly reduced *CX3CR1* mRNA, a chemokine receptor ( $p=0.0345$ ; Fig. 7H). TAM-treated *APP*;C3iKO mice showed no change in *Tgfr1* or *CR3* (*CD18*, *CD11b*) mRNA levels, indicating that *C3* lowering had no effect on *Tgfr1* and *CR3* gene expression (Fig. 7I-L).

### **C3 lowering protected against the loss of synaptic proteins and maintained synaptic integrity in aged *APP*;C3iKO mice**

We conducted immunofluorescence staining for synaptic markers in the hippocampal CA1 and CA3 regions to determine the effects of *C3* reduction in aged *APP*;C3iKO mice. We specifically examined the expression of pre-synaptic (synaptophysin, SYN, and Bassoon, BSN) and postsynaptic (PSD95 and Homer1) proteins and their colocalization in synaptic puncta to evaluate synaptic integrity. Higher-magnification confocal images (Fig. 8A) were quantified for presynaptic and postsynaptic markers (Fig. 8B-M).

In the hippocampal CA1 region of TAM-treated *APP*;C3iKO mice, there was a significantly higher density of PSD95 (Fig. 8B) and SYN (Fig. 8C) synapse labeling compared to CO-treated *APP*;C3iKO mice.

Increased colocalization of PSD95 and SYN within synaptic puncta in CA1 confirmed neuroprotective effects of *C3* lowering (Fig. 8A, 8D). Similar results were seen in hippocampal CA3 in which PSD (Fig. 8F) and SYP (Fig. 8E) synaptic densities and colocalization of PSD95/SYP in synaptic puncta (Fig. 8G) were significantly higher in TAM-treated *APP*;C3iKO mice compared to CO-treated controls. Furthermore, presynaptic marker BSN and postsynaptic marker Homer1 individual synaptic densities and colocalization in synaptic puncta were higher in CA1 (Fig. 8H-J) and CA3 (Fig. 8K-M) in TAM-treated *APP*;C3iKO mice compared to CO-treated controls. Furthermore, both the quantitative and qualitative data showing the colocalization of *C3* with the presynaptic protein Bassoon confirm that the reduction of *C3* plays an important role in preserving synaptic proteins in TAM-treated *APP*;C3iKO mice (Fig. 9A-B). Taken together, these results provide evidence that *C3* lowering early in plaque deposition, provided protection of hippocampal synapses with aging and further Alzheimer's pathogenesis in TAM-treated *APP*;C3iKO mice.

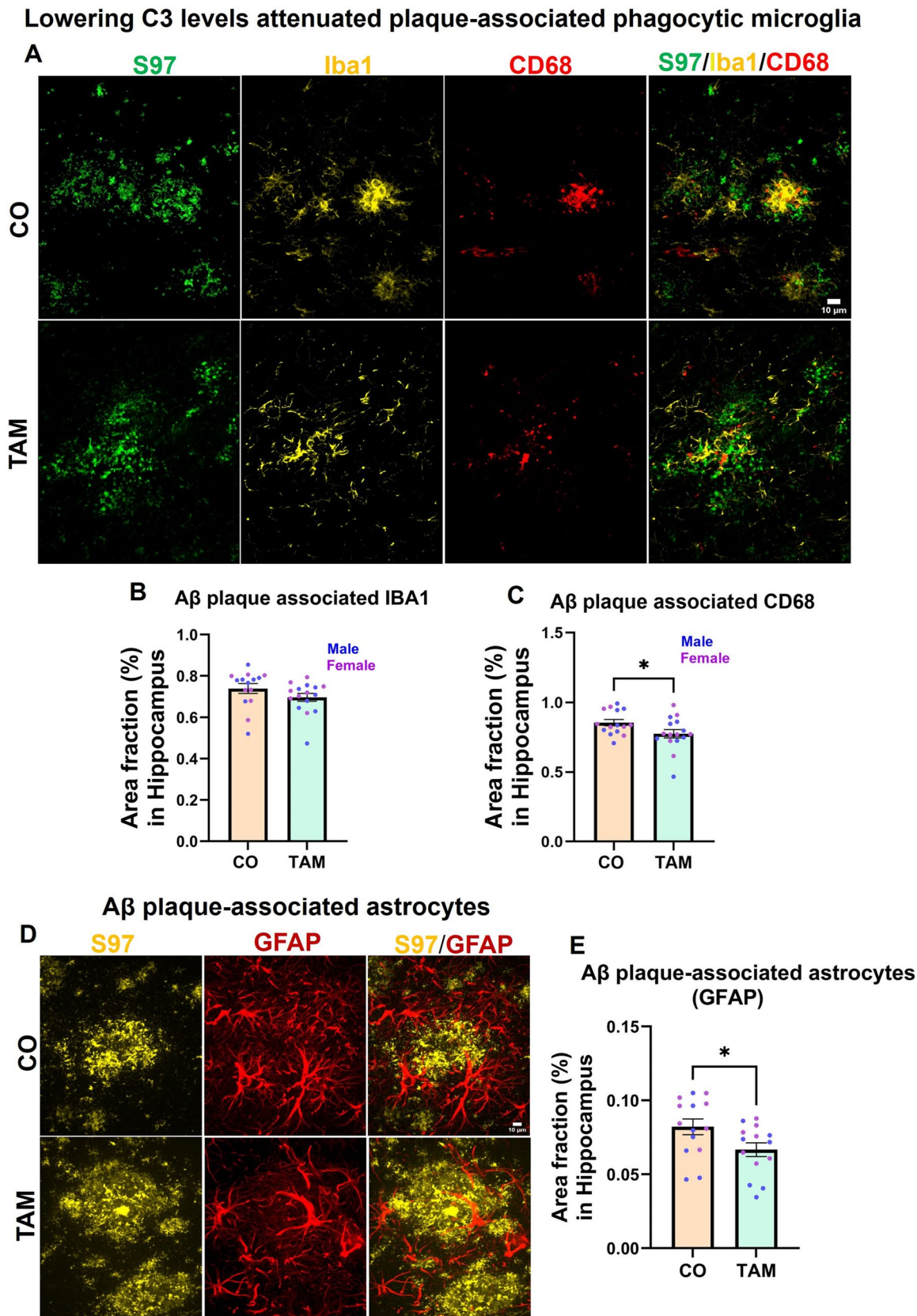
### **C3 lowering differentially upregulated synaptic signaling-related genes in aged *APP*;C3iKO mice**

We conducted bulk RNA sequencing on the prefrontal cortex of TAM- and CO-treated *APP*;C3iKO mice and identified 1,071 differentially expressed genes (DEGs). Among these, 569 genes were upregulated, and 502 genes were downregulated compared to those in CO-injected *APP*;C3iKO mice. To visualize these expression changes, we generated a heat map (Fig. 10A), and a volcano plot (Fig. 10B) based on the top-most significant fold-change and low  $p$  values, focusing on the complement component and synaptic genes.

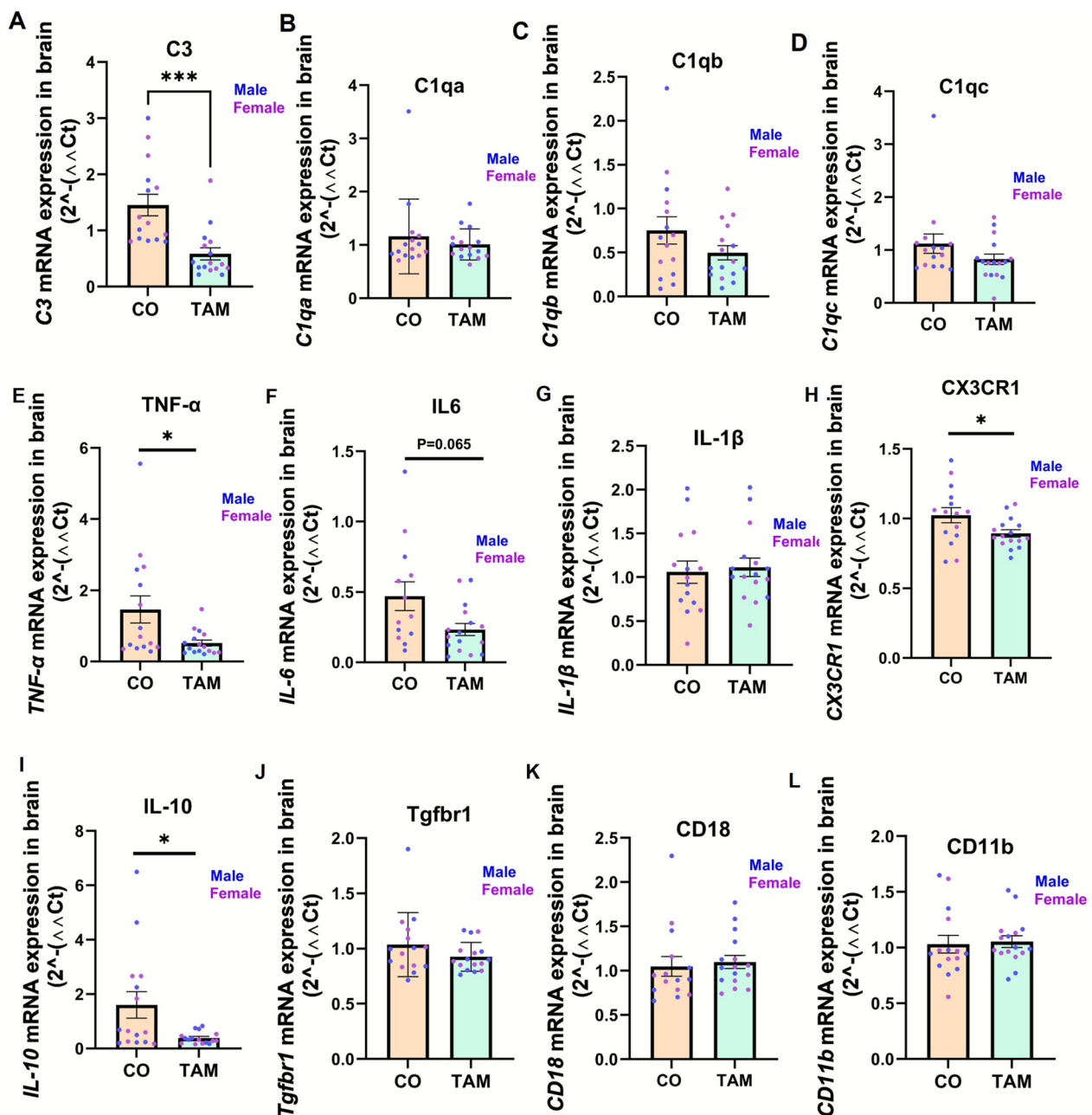
The expression levels of the top DEGs are highlighted in the heat map, where red and blue denote upregulation and downregulation, respectively. Samples from TAM-treated mice exhibited distinct expression profiles compared to CO-treated control mice. As expected, we found that *C3* expression, but not *C1q* expression, was significantly downregulated in *APP*;C3iKO mice, confirming the successful lowering of *C3*. Additionally, we found that among highest

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**Fig. 6** A $\beta$  plaque-associated phagocytic microglia (CD68) and astrocytes (GFAP) in 15–16 month-old *APP*;C3iKO mice. (A) Representative confocal triple-immunofluorescence images showing S97 (A $\beta$ , green), IBA1 (microglia, yellow), and CD68 (phagocytic marker, red) immunostaining in the hippocampus of *APP*;C3iKO mice. (B) Quantitative analysis of A $\beta$  plaque-associated IBA1 + microglia. (C) Quantitative analysis of A $\beta$  plaque-associated CD68 + phagocytic microglia. N = 15–16 mice per group. (D) Representative confocal double-immunofluorescence images showing S97 (A $\beta$ , yellow) and GFAP + astrocytes (red) in the hippocampus of *APP*;C3iKO mice. (E) Quantitative analysis of A $\beta$  plaque (S97)-associated GFAP + astrocytes. N = 14 mice per group. Scale bar = 100  $\mu$ m. Data are presented as mean  $\pm$  SEM. Statistical analysis was performed using pairwise comparisons with Student's  $t$ -test.  $P$  values less than 0.05 (\* $p < 0.05$ ) were considered significant



**Fig. 6** (See legend on previous page.)



**Fig. 7** Immune gene expression (qRT-PCR) in brain in 15–16 month-old mice. (A) C3 mRNA expression. (B) C1qa mRNA expression. (C) C1qb mRNA expression. (D) C1qc mRNA expression. (E) TNF- $\alpha$  mRNA expression. (F) IL-6 mRNA expression. (G) IL-1 $\beta$  mRNA expression. (H) CX3CR1 mRNA expression. (I) IL-10 mRNA expression. (J) Tgfb $\beta$ 1 mRNA expression. (K) CD18 mRNA expression. (L) CD11b mRNA expression. Data are presented as mean  $\pm$  SEM. N = 13–16 mice per group. Statistical analysis was performed using pairwise comparisons with Student's t-test. P values less than 0.05 (\*p < 0.05) and 0.001 (\*\*\*p < 0.001) were considered significant

top-ranking upregulated DEGs, *Bsn*, *Homer1*, *Synapsin II* (*Syn2*), *Synaptopodin* (*SYNPO*), and *SNAP25* are linked to synaptic signaling processes and essential for synaptic structure and function. Increased expression of these genes in the prefrontal cortex of TAM-treated

*APP*;C3iKO mice is consistent with strong synaptic protection.

## Discussion

To address the effect of C3 lowering after the onset of plaque deposition on the pathophysiology of Alzheimer's disease (AD), we generated an inducible conditional *APP*;C3iKO mouse model by crossing *APP<sup>NL-G-F/NL-G-F</sup>* mice with C3iKO mice. Following TAM injections at ~3.6 months of age, after the onset of plaque deposition, we confirmed the efficacy of global C3 lowering in *APP*;C3iKO mice had a sustained, significant lowering of C3 in serum and brain at 15–16 months of age, confirming the effectiveness of global complement C3 reduction over time in *APP*;C3iKO mice. However, we observed a reduction in serum C3 levels (Fig. 1C and D) with age which may be due to an uptake of peripheral C3 into the CNS, deposition into cerebral amyloid plaques and/or increased cleavage of C3 into C3a and C3b. This hypothesis is supported by the fact that the blood–brain barrier (BBB) is characterized by both aging and amyloid pathology [30], and BBB dysfunction early in AD pathogenesis in the *APP*-KI mouse model [31].

Global reduction of complement C3 after the onset of amyloid plaque formation mitigated cognitive decline in our aged *APP*;C3iKO mice but had no significant effects on locomotion or anxiety. This is consistent with our earlier results showing that lifelong germline C3 knockout in *APP*/PS1 mice protected against cognitive decline but had only minimal effects on locomotion and anxiety [22]. Here, no significant effects on C3 lowering were seen in the Open Field test except for modest trends for less hyperactivity and anxiety, consistent with our earlier study [22] and a previous report that showed that reduced neuroinflammation was associated with less anxiety-like behavior [32]. Importantly, TAM-treated *APP*;C3iKO mice performed significantly better in all three assessments of cognitive function, including the Novel Object Recognition (NOR), Novel Object Location (NOL), and Spatial Novel Y-Maze (SNYM) tests. In the SNYM test, TAM-treated *APP*;C3iKO mice had better spatial working memory, as evidenced by spending more time in the novel arm than CO-treated mice. Given that AD is characterized by deficits in spatial memory, this finding is significant because improvements in this area indicate a significant cognitive benefit from C3

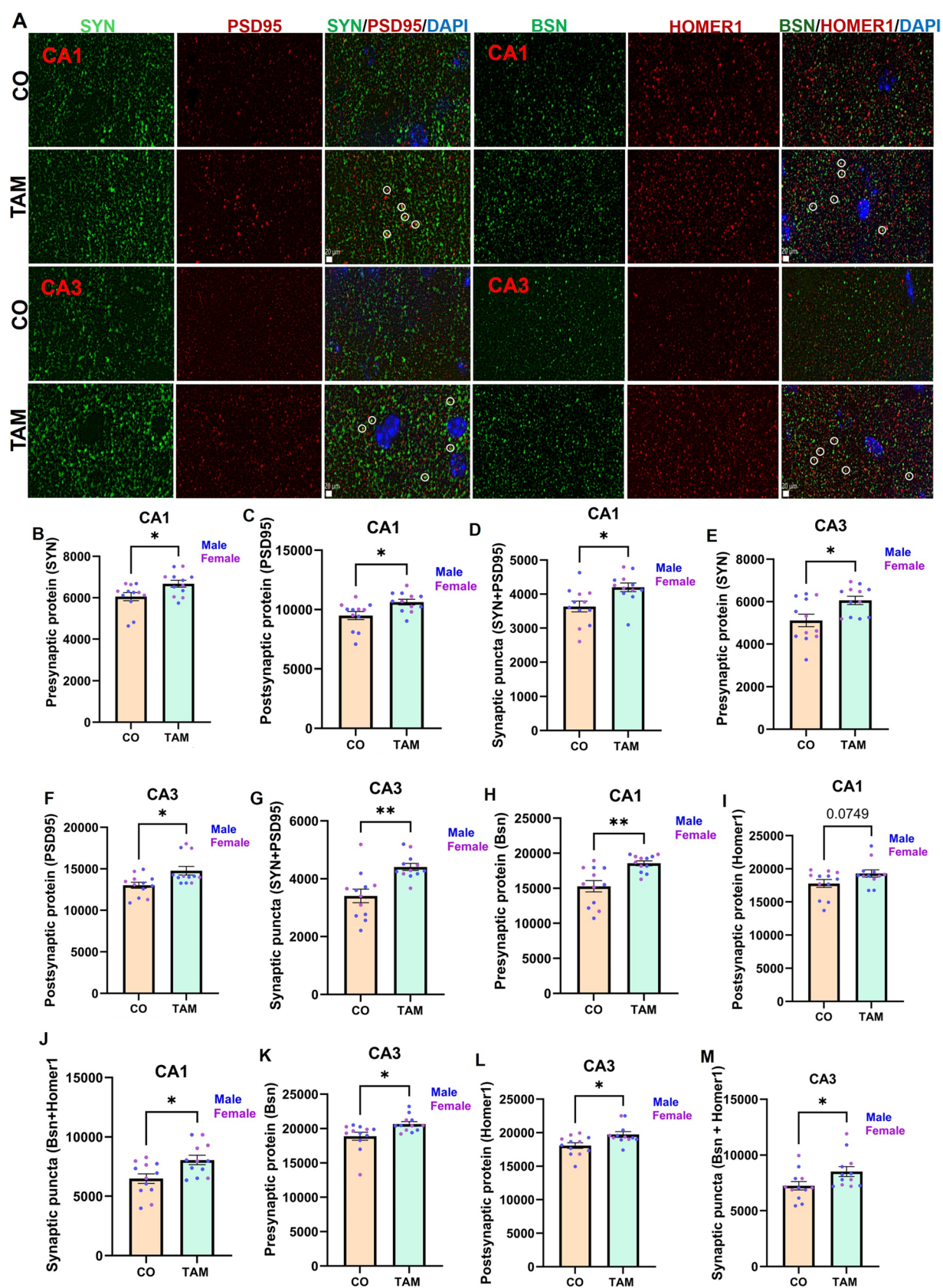
suppression [33]. Similarly, in the NOR and NOL tests, TAM-treated mice showed enhanced recognition and location memory, as indicated by a higher discrimination index and preferential exploration of the novel object and location, indicated cognitive preservation due to C3 lowering in *APP*;C3iKO mice. Together, these results demonstrate that C3 lowering confers cognitive protection in *APP*;C3iKO aged mice, possibly by reducing neuroinflammation [34] and maintaining synaptic integrity [35], as C3 activation has been linked to neurodegeneration and synapse loss [36, 37].

Our study of the long-term consequences of reducing complement C3 in *APP*;C3iKO mice contributes to the understanding of complement on cellular and molecular changes, particularly in relation to Alzheimer's disease (AD). C3 lowering in *APP*;C3iKO mice led to a significant decrease in C1q protein levels, which is another important protein in the complement cascade. A significant reduction in C1q fluorescence intensity in the hippocampus region and C1q levels in the T-per brain homogenate revealed the impact of C3 lowering on other complement proteins in the brains of aged *APP*;C3iKO mice. Although *C1q* gene expression in brain was not significantly altered by C3 lowering, there was a trend for a reduction, consistent with the results in our companion paper (Batista et al.) in which *C1qa/b/c* mRNA and C1q protein were significantly reduced in aged C3iKO mouse brain after a year of C3 lowering. We did not examine C1q or other complement components in our earlier papers [22, 38] thus this adds to the novelty of this paper. Giesbrecht et al. (2022) demonstrated that targeting C3 deficiency can lead to reductions in other complement proteins [39] thereby attenuating complement-mediated neuroinflammation and neurodegeneration.

Considerable A $\beta$  buildup and neurodegeneration have been reported in the hippocampus, making it an area highly affected in AD pathogenesis [40, 41]. This region is important for cognitive functions such as learning and memory, which are impaired in Alzheimer's disease patients [42]. Therefore, we focused on the hippocampus in this study. Despite the sparing of cognitive decline, C3 lowering in adulthood had no effect on Ab plaque load in the hippocampus of TAM-treated *APP*;C3iKO mice

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**Fig. 8** Pre- and post-synaptic staining and colocalization of synaptic puncta in 15–16 month-old mice. A) Representative confocal double-immunolabeling images showing Synaptophysin (SYN), PSD95, Bassoon, and Homer1 in the CA3 and CA1 hippocampal subregions. Blue staining represents DAPI, used to label cell nuclei. White circles indicate representative synaptic puncta. Synaptic puncta were defined as regions where pre- and postsynaptic markers (SYN + PSD95 or Bassoon + Homer1) colocalize or are in close spatial proximity within a defined threshold as defined in the Methods. (B–G) Quantitative analysis of synaptic puncta for SYN, PSD95, and SYN + PSD95 synaptic puncta. (H–M) Quantitative analysis of synaptic puncta for Bassoon, Homer1, and Bassoon + Homer1 synaptic puncta. Scale bar = 20  $\mu$ m. Data are presented as mean  $\pm$  SEM. N = 12 mice per group. Statistical analysis was performed using pairwise comparisons with Student's t-test. \* $p \leq 0.05$ , \*\* $p \leq 0.01$  were considered significant



**Fig. 8** (See legend on previous page.)

expressing physiological levels of Ab. This result is in line with the parallel study by Dujardin et al. (Vandenbroucke lab, unpublished data) in an accompanying paper in which C3 lowering in adult *APP;C3iKO* mice had no effect on cerebral amyloid burden at 8–10 months of age. In contrast, we and others previously reported that germline C3 depletion in 3 different transgenic mouse models of amyloidosis (J20, *APP/PS1dE9* and *PSAPP*) caused an increase in plaque deposition, possibly due to the strong overexpression of human mutant APP in these models [22, 43]. C3 lowering spared synaptic loss in the *APP/PS1* models, despite increased plaque deposition [22], but worsened it in the J20 model [43], possibly due to differences in the glial immune response between the models. Furthermore, Dodart et al. (2002) showed that, while improving behavioral outcomes, altering immune responses in murine amyloid models of AD did not consistently reduce the number of A $\beta$  plaques, suggesting that C3 plays a multifaceted role in amyloid pathology [44]. The inability of C3 lowering to alter the amyloid plaque load in *APP;C3iKO* mice highlights the intricacy of AD pathogenesis and raises the possibility that complement modulation predominantly benefits the brain through mechanisms other than direct plaque removal. Instead, we speculate that C3 lowering reduced neuroinflammation and protected synapses, both of which likely contributed to sparing of cognitive decline as described below.

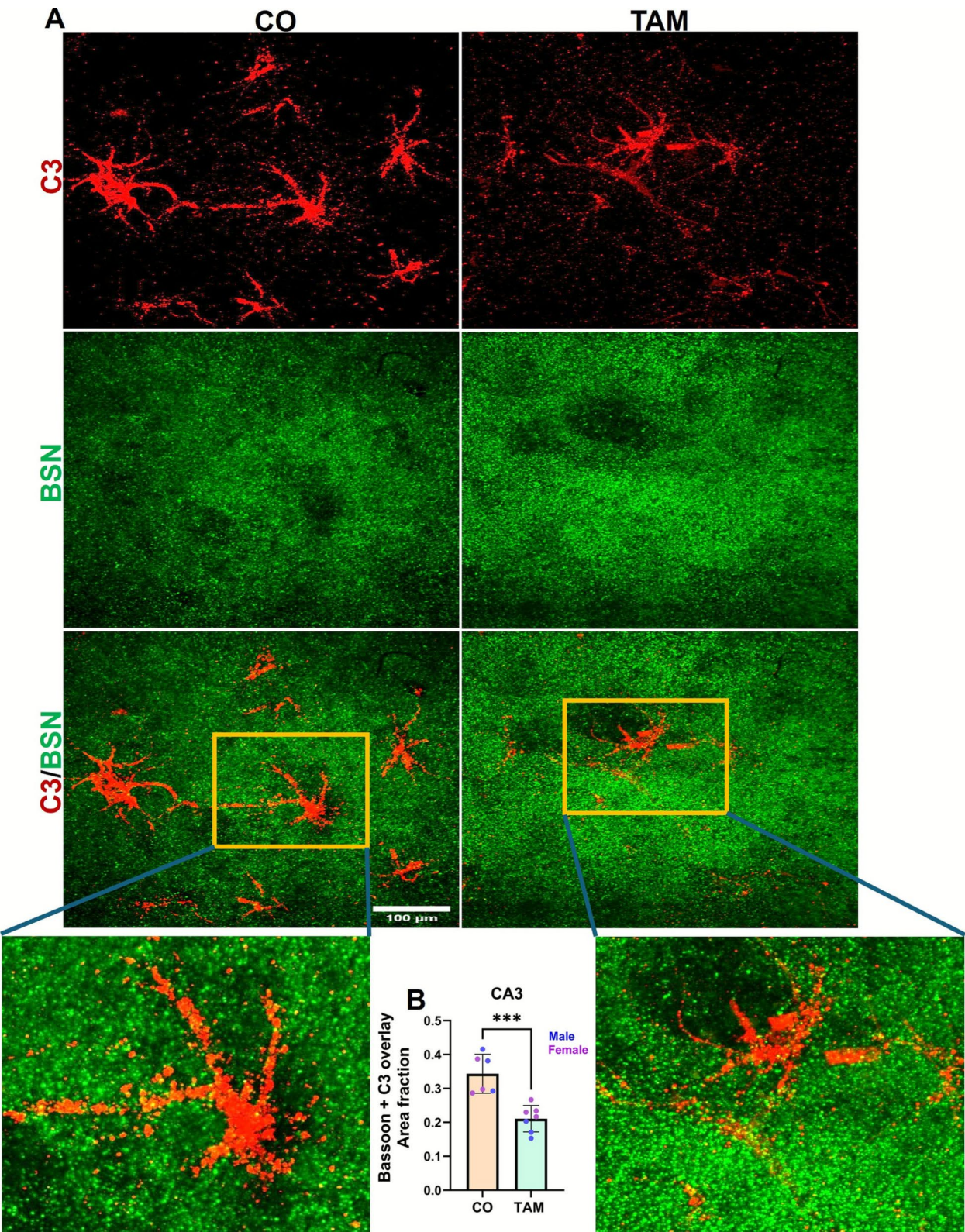
While modest overall reductions in hippocampal microglial labeling, especially in CA3, and no changes in astrocytic GFAP were seen with C3 lowering in *APP;C3iKO* mice, significant reductions in plaque-associated phagocytic microglia and plaque-associated astrocytes were observed. Spatial memory formation and maintenance depend mostly on CA3, but in AD, synapse loss and neuroinflammation can cause significant alterations in the hippocampal CA3 subregion [45–47]. Our study aligns with previous studies that demonstrated the vulnerability of CA3 subregion to pathological alterations with aging and AD [48, 49], and suggests that reduced microglial activity may attenuate the neuroinflammatory response, thereby maintaining the critical functions of the CA3 region and preserving cognition. Furthermore, no overall discernible differences were found in GFAP immunoreactivity in CO- and TAM-treated *APP;C3iKO*

mice, indicating that C3 lowering did not broadly impact overall astrocytic activation. However, we observed a localized reduction in astrocytes associated with plaques in the hippocampus following C3 lowering in *APP;C3iKO* mice. This finding is similar to our earlier study in germline C3-deficient *APP/PS1* mice in which there was no overall reduction in hippocampal astrocytes, but there were fewer microglia and astrocytes found infiltrating in the center of the plaques [22]. This highlights the complex role of astrocytes in the brain of mouse models of amyloidosis. Although astrocytes are the main source of C3 in the brain, they appear to be less directly affected by C3 lowering than microglia, which can also produce C3. Microglia are vital for both phagocytosis of A $\beta$  plaques and neuroinflammatory responses [50, 51]. Our finding that C3 lowering in adulthood reduced CD68+ phagocytic microglia associated with hippocampal plaques with aging is in line with our earlier study in *APP/PS1;C3iKO* mice [22] and other studies that have shown that decreased neuroinflammatory changes in the microglial response towards amyloid plaques [52, 53]. The interplay between A $\beta$  and astrocytes is a well-known contributing factor to neurotoxicity in A $\beta$ -associated diseases [54], despite the protective and neuronal supporting roles of astrocytes [55]. In AD, C3 is produced by astrocytes in response to A $\beta$  plaques, and its cleaved component C3a binds to the C3aR receptor on microglia, inducing microglial activation and subsequent neuroinflammation [17, 56].

Neurodegeneration and cognitive impairment are associated with persistent inflammation [57]. During the phagocytic engulfment of A $\beta$  plaques, highly activated microglia secrete proinflammatory cytokines such as TNF- $\alpha$ , IL-1 $\beta$ , and IL-6 [58]. These cytokines may result in a long-lasting neuroinflammatory reaction that compromises synaptic function and promotes the growth of neurotoxic environment in the brain [59, 60]. In neuroinflammatory states, TNF $\alpha$  plays a central role as an inflammatory mediator in neurodegenerative diseases such as AD [61]. There is substantial evidence for elevated levels of proinflammatory cytokines (e.g., TNF $\alpha$ , IL-6, and IL-1 $\beta$ ) in CSF and brain of AD patients compared to non-AD control brains [62]. Here, we found significant reductions in the mRNA expression of the pro-inflammatory cytokine *TNF $\alpha$*  in TAM-injected *APP;C3iKO*

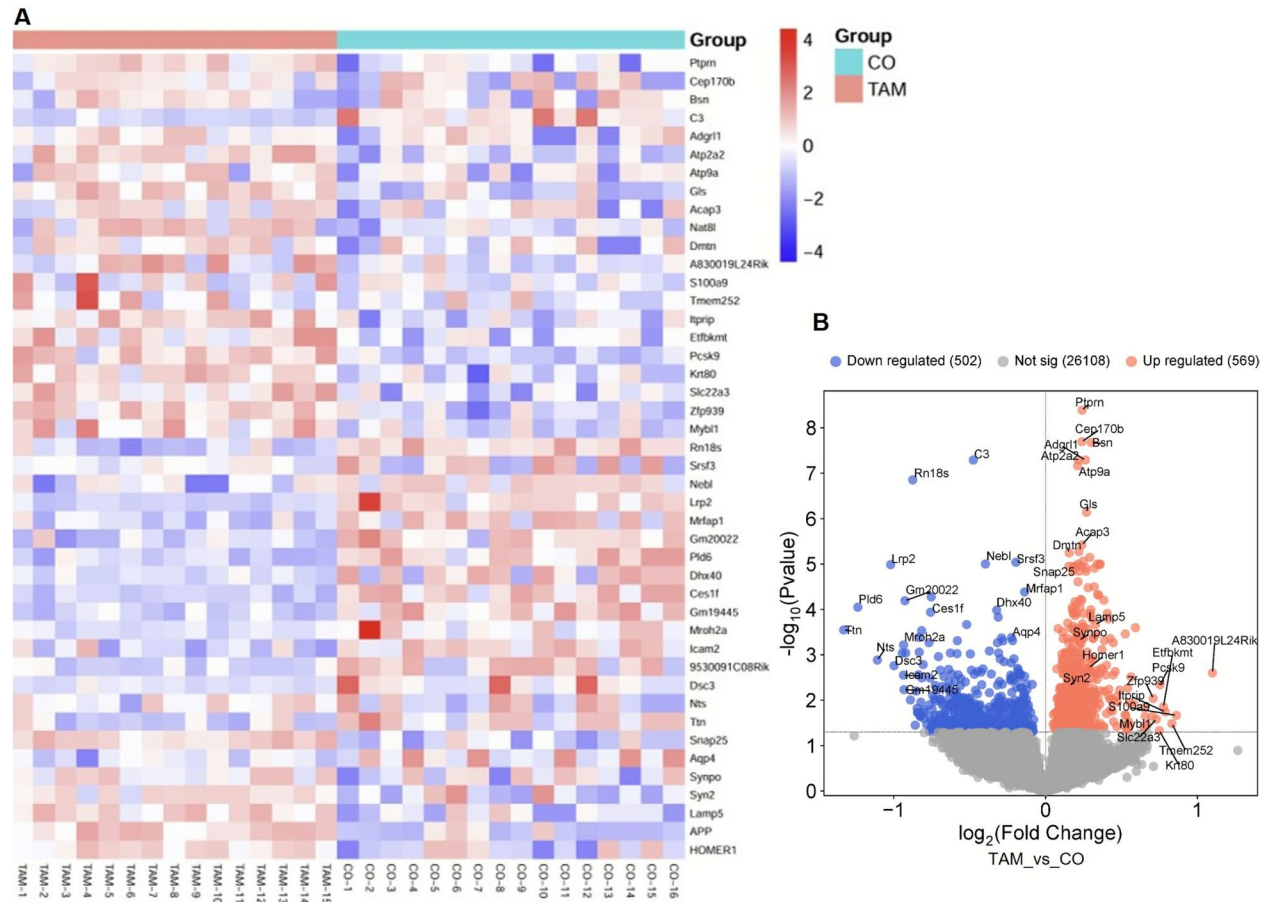
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**Fig. 9** C3 tagged-Bassoon in TAM-injected *APP;C3iKO* mice. (A) Representative confocal images showing co-localization of C3 (red) and Bassoon (green) in the CA3 region of the hippocampus in 15–16-month-old *APP;C3iKO* mice injected with corn oil (CO) or tamoxifen (TAM). Co-localization of Bassoon and C3 overlay was quantified using an ImageJ macro to ensure objective and consistent analysis. Scale bar = 100  $\mu$ m. (B) Quantitative analysis of C3-tagged Bassoon presynaptic elements in the CA3 subregion of TAM-injected *APP;C3iKO* mice. Data are presented as mean  $\pm$  SEM. N=6–7 mice per group. Statistical analysis was performed using pairwise comparisons with Student's t-test. \*\*\* $p \leq 0.001$  was considered significant



**Fig. 9** (See legend on previous page.)

Bulk RNA Seq



**Fig. 10** Unbiased bulk RNAseq of pre-frontal cortex in 15–16 month-old *APP;C3iKO* mice. (A) 1071 differentially expressed genes (DEGs) were identified in which 569 upregulated genes and 502 down-regulated genes in TAM-injected *APP;C3iKO* mice with the top hits shown in the HEAT map. (B) A volcano plot illustrates the most significantly altered genes in the TAM-injected vs. CO-injected *APP;C3iKO* mice by fold-change and low p value. N = 15–16 mice per group. The majority of differentially expressed genes (DEGs) that rank highest are associated with synaptic signaling functions (Bassoon, Homer1, Syn2, SYNPO, and SNAP25)

mice, suggesting an attenuated neuroinflammatory state, which may be associated with cognitive sparing in these mice. Consistent with lower *TNFα* expression, we found a strong trend for reduced *IL-6* mRNA levels upon C3 lowering, confirming a less inflammatory microenvironment in *APP;C3iKO* mice. Surprisingly, *IL-10* mRNA levels were also reduced with C3 lowering suggesting suppression of this anti-inflammatory cytokine. While the expression levels of *IL-1β*, *Tgfbr1*, *CR3* (*CD18* and *CD11b*) in TAM-injected *APP;C3iKO* mice remained unchanged, it is possible that C3 lowering only affects selected genes. Additionally, C3 lowering resulted in reduced *CX3CR1* mRNA, a chemokine receptor

implicated in immune cell recruitment [63]. Interestingly, *CX3CR1*-deficiency reduced microglial activation and plaque-associated microglia in the *APP<sup>PS1</sup>* mouse model of AD [64].

In AD, microgliosis and enhanced phagocytosis of synaptic components and plaques contribute to synaptic deterioration [65]. Moreover, C3 is involved in synaptic pruning during development, synaptic degeneration during aging, and neuroinflammatory conditions [10]. Therefore, we asked whether the reduced inflammatory state due to C3 lowering in *APP;C3iKO* mice protected against hippocampal synaptic loss. We found that C3 lowering increased hippocampal pre- and post-synaptic

markers as well as colocalized synaptic puncta in TAM-treated *APP*;C3iKO mice compared to control mice treated with corn oil, conferring synaptic protection. Our findings are consistent with those of our previous study and another study showing that germline C3-deficiency protected synaptic integrity in *APP/PS1* and *PSAPP* transgenic mice [22]. Here, we expanded our previous work by performing bulk RNA sequencing of tissue from the prefrontal cortex, a brain region crucial for higher-order cognitive processes including working memory, executive function, and decision-making, as well as one of the regions in AD that is most susceptible to synaptic loss and dysfunction [66, 67]. Specifically, 1071 differentially expressed genes (DEGs) were found in the prefrontal cortex of TAM- vs. CO-treated *APP*;C3iKO mice. Among these, TAM-treated mice showed upregulation of 569 genes and downregulation of 502 genes compared with CO-treated mice. Many synaptic genes including *SNAP25*, *Homer1*, *Synapsin II* (*Syn2*), *Bassoon* (*Bsn*), and *Synaptopodin* (*Synpo*), were significantly upregulated, consistent with synaptic preservation and repair in the brains of TAM-treated and CO-treated *APP*;C3iKO mice. Bassoon and Homer-1 are essential for synaptic vesicle release and postsynaptic signaling in the brain [68, 69]. Neurotransmitter release is dependent on synapsinII and SNAP25, whereas synaptopodin plays a role in spinal apparatus formation and synaptic plasticity [70, 71]. Synaptic genes were upregulated in TAM-treated *APP*;C3iKO mice showing that C3 lowering may enhance synaptic resilience and adaptation in addition to protecting existing synapses.

Our study presents novel findings in terms of the timing of global C3 suppression and the different mouse model applied—a knock-in mouse model that ensures APP is expressed at physiological levels, minimizing confounds [23]. Previous studies crossed germline C3 knockout to *APP/PS1* and J20 mice models which demonstrated that lifelong C3 deficiency provided cognitive and synaptic protection despite an increase in plaque burden in aged mice [13, 22, 43]. In contrast, in this study, we induced the C3 lowering in early adulthood after the onset of amyloid deposition, rather than prior to brain development. However, we did not observe a complete C3 deficiency in the *APP*-KI mice brain which is a major limitation of this study. This may explain why we did not see major glial changes in the whole hippocampus, except for the changes in glial association within and near plaques. Similarly, *APP/PS1*;C3KO mice showed no changes in microglia or astrocyte numbers and a significant decrease in microglia and astrocytes migrating into the center of plaques in the hippocampus region [22]. However, a recent publication by Dujardin et al. showed that while C3 expression in microglia and astrocytes

increased with age in *APPNL-G-F* KI mice, near-complete C3 lowering by tamoxifen treatment did not affect gliosis or amyloid burden [72]. Nevertheless, even without the complete depletion of C3, we found hippocampal synaptic protection in our aged *APP*;C3iKO mice following global C3 lowering in early adulthood brains after the early amyloid deposition.

Some of the limitations of this study are the absence of age-matched wild-type control mice, which prevents directly assessing how much C3 lowering in APP-KI mice mitigated the synaptic deficits and cognitive decline compared to mice lacking amyloid plaques. Notably, the broad approach of use of global C3 lowering in adulthood may be informative in animal models, but it is unlikely that systemic C3 lowering in AD patients would be safe over the long term, as it would leave them vulnerable to infections. Instead, brain-specific reduction of C3 or its downstream components of C3 activation (such as C3aR, C5aR1, C7 or MAC) may be more favorable as a treatment to protect synapses. For example, Carpanini (2022) showed that blocking membrane attack complex (MAC) formation through systemic administration of an anti-C7 antibody in *APP<sup>NL-G-F</sup>* KI mice reduced hippocampal synapse loss [73]. They also reported that C6 deficiency in 3xTg-AD mice rescued hippocampal synapses and spine density. Zelek (2024) demonstrated that C7 deficiency in *APP<sup>NL-G-F</sup>* KI mice reduced MAC formation, hippocampal synapse loss and amyloid plaque burden, and protected against cognitive decline [74]. While still in its early days, targeting of complement in neurodegenerative diseases is an evolving and fast-growing research area.

## Conclusions

Our study demonstrated that C3 lowering after the onset of amyloid deposition (~3.6 months of age) in *APP*;C3iKO mice mitigated cognitive decline, lowered neuroinflammation, upregulated synaptic genes, and preserved synaptic integrity in aged *APP*;C3iKO mice, without affecting cerebral amyloid plaque load, suggesting that C3 lowering, especially in the brain, early in cerebral Ab accumulation could be a viable therapeutic strategy for Alzheimer's disease (AD).

## Abbreviations

|                             |                                                                      |
|-----------------------------|----------------------------------------------------------------------|
| <b>Aβ</b>                   | Amyloid-beta                                                         |
| <b>AD</b>                   | Alzheimer's Disease                                                  |
| <b>APP</b>                  | Amyloid Precursor Protein                                            |
| <b>APP-KI</b>               | APP Knock-In                                                         |
| <b>APP;C3iKO</b>            | APP <sup>NL-G-F</sup> ;C3 Inducible Knockout                         |
| <b>APP/PS1</b>              | APP Presenilin-1 Transgenic Mouse Model                              |
| <b>APP<sup>NL-G-F</sup></b> | APP Knock-in Mouse Model with Swedish, Arctic, and Iberian Mutations |
| <b>BSA</b>                  | Bovine Serum Albumin                                                 |
| <b>BSN</b>                  | Bassoon (Presynaptic Marker)                                         |
| <b>CA1, CA3</b>             | Cornu Ammonis Subregions of the Hippocampus                          |

|                               |                                                                |
|-------------------------------|----------------------------------------------------------------|
| <b>C1q</b>                    | Complement Component 1q                                        |
| <b>C3</b>                     | Complement Component 3                                         |
| <b>C3KO</b>                   | Complement Component 3 Knockout                                |
| <b>C3iKO</b>                  | Inducible Conditional Knockout of Complement Component 3       |
| <b>C3aR</b>                   | Complement Component 3a Receptor                               |
| <b>CD11b, CD18</b>            | Components of Complement Receptor 3 (CR3)                      |
| <b>CD68</b>                   | Phagocytic Microglial Marker                                   |
| <b>CO</b>                     | Corn Oil                                                       |
| <b>Cre</b>                    | Cre Recombinase                                                |
| <b>CR3</b>                    | Complement Receptor 3                                          |
| <b>CR3CR1</b>                 | Fractalkine Receptor                                           |
| <b>DAB</b>                    | 3,3'-Diaminobenzidine                                          |
| <b>DEGs</b>                   | Differentially Expressed Genes                                 |
| <b>DG</b>                     | Dentate Gyrus                                                  |
| <b>ELISA</b>                  | Enzyme-Linked Immunosorbent Assay                              |
| <b>GFAP</b>                   | Glial Fibrillary Acidic Protein (Astrocyte Marker)             |
| <b>GuHCl</b>                  | Guanidine Hydrochloride                                        |
| <b>IBA1</b>                   | Ionized Calcium-Binding Adapter Molecule 1 (Microglial Marker) |
| <b>IACUC</b>                  | Institutional Animal Care and Use Committee                    |
| <b>IL-1<math>\beta</math></b> | Interleukin-1 Beta                                             |
| <b>IL-6</b>                   | Interleukin-6                                                  |
| <b>IL-10</b>                  | Interleukin-10                                                 |
| <b>ITI</b>                    | Intertrial Interval                                            |
| <b>MAC</b>                    | Membrane Attack Complex                                        |
| <b>mRNA</b>                   | Messenger Ribonucleic Acid                                     |
| <b>MSD</b>                    | Meso Scale Discovery                                           |
| <b>NOL</b>                    | Novel Object Location Test                                     |
| <b>NOR</b>                    | Novel Object Recognition Test                                  |
| <b>OF</b>                     | Open Field Test                                                |
| <b>PBS</b>                    | Phosphate-Buffered Saline                                      |
| <b>PBS-T</b>                  | Phosphate-Buffered Saline with Tween                           |
| <b>PSD-95</b>                 | Postsynaptic Density Protein 95                                |
| <b>qPCR</b>                   | Quantitative Polymerase Chain Reaction                         |
| <b>RNA-seq</b>                | RNA Sequencing                                                 |
| <b>SNYM</b>                   | Spatial Novel Y-Maze Test                                      |
| <b>Syn2</b>                   | Synapsin II                                                    |
| <b>SYN</b>                    | Synaptophysin (Presynaptic Marker)                             |
| <b>SNAP25</b>                 | Synaptosomal-Associated Protein 25                             |
| <b>TAM</b>                    | Tamoxifen                                                      |
| <b>Tgfr1</b>                  | Transforming Growth Factor Beta Receptor 1                     |
| <b>ThioS</b>                  | Thioflavin S                                                   |
| <b>TNF<math>\alpha</math></b> | Tumor Necrosis Factor Alpha                                    |
| <b>WT</b>                     | Wild-type                                                      |

## Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s44477-025-00009-6>.

Below is the link to the electronic supplementary material. Supplementary file1 (PDF 274 KB)

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## Author contributions

CAL, TCS and MCC designed and planned the studies. TCS provided the \*APP-KI\* mice. BS, AFB, ETS and BRC performed the experiments. BS, ETS, BRC and CAL analyzed and interpreted the data. BS and CAL prepared the manuscript. All authors reviewed the manuscript.

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## Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

## Declarations

### Ethical approval

The Institutional Animal Care and Use Committee (IACUC) of Brigham and Women's Hospital approved the experimental procedures, which followed protocols aimed at minimizing animal suffering and the number of animals utilized (BWH Protocol: 2,016,000,396; CAL).

### Consent for publication

Not applicable.

### Competing interests

CAL serves as a consultant to and has an unrelated sponsored research agreement from Apellis Pharmaceuticals. All other coauthors declare they have no competing interests.

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